

SAMPLE LESSON · CASE STUDY: POTATO LATE BLIGHT

Can We Out-Breed Crop Disease?

A look inside a World Science Academy research session — where a real plant-disease problem becomes award-winning research.



TODAY'S CHALLENGE

A single disease once starved a nation.

Can we breed crops that resist it — and keep that resistance from breaking?



The Great Famine — Ireland, 1845–1852. Late blight destroyed the potato crop; over a million people died.

BY THE END, STUDENTS CAN

- ✓ Compare the tradeoffs of different disease-resistance strategies
- ✓ Explain why pathogens evolve past single-gene resistance
- ✓ Read real yield-and-cost data to make a recommendation

THE THREAT

Meet Late Blight

Caused by **Phytophthora infestans** — a fast-spreading, fungus-like water mould that can wipe out a whole field within days.

01 Spreads fast

Wind-blown spores jump field to field in cool, wet weather.

02 Devastating

Triggered the Irish Potato Famine (1845–1852).

03 A founding case

Its cause, proven in 1876, gave birth to plant pathology.



Blackened leaves & stems in the field



Brown rot spreading inside the tuber

Why a single defence fails

Variation

Mutations create new pathogen types.

Selection

A resistant crop kills off the susceptible ones.

Reproduction

Survivors multiply fast, spreading their genes.

Breakdown

The crop's single defence stops working.

The faster a pathogen varies and reproduces, the faster it evolves past a crop's defences.

KEY CONCEPTS

The language students build

Disease resistance

A plant variety's ability to limit a pathogen's growth and the damage it causes.

Pathogen

An organism that causes disease.

Natural selection

Individuals best adapted to their environment survive and reproduce.

Students do, not just listen

In-Person

- Real-time polling & word clouds
- Think-Pair-Share discussions
- Live dataset & graph analysis
- Case-study debates

Online

- Social annotation of case studies
- Micro-module self-checks
- Small-group discussion pods
- Scenario-based simulations

Which resistance strategy holds up?

Students compare four ways to defend a potato crop — and the trade-offs behind each.

Strategy	Yield loss (light)	Yield loss (outbreak)	Durability
Single R-gene	3%	52%	Low
Stacked R-genes	4%	8%	High
Variety mixture	14%	16%	Moderate
Susceptible + fungicide	6%	28%	High*

*Depends on continued spraying and no fungicide resistance. Illustrative teaching figures.

THE BIG IDEA

There is no single fix — only trade-offs

A single strong gene looks perfect until the pathogen evolves. Durable protection comes from **stacking defences** and **diversity** — the same logic behind vaccines, antibiotics, and crop breeding alike.

Evolution never stops

Diversity buys durability

Every choice has a cost

Crop vs. pathogen, through history

- 1845 **Irish Potato Famine.** Late blight destroys the crop; over a million die.
- 1876 **de Bary** proves a microbe — *P. infestans* — causes it. Plant pathology is born.
- 1900s Breeders move **R-genes** from wild potato into crop varieties.
- 1950s New pathogen races **overcome single R-genes** — resistance breaks down.
- Today** **Gene stacking & diversity** aim for durable, longer-lasting resistance.



Turning this into a science-fair study

Ask a testable question

- Do mixed varieties slow disease vs. one variety?
- Which leaf traits predict resistance?
- How does temperature change spread rate?

Design a fair test

- Controls, replicates & a clear variable
- Measure & score disease objectively
- Graph the data — then argue from it

Our role: we coach students through each step — framing the question, running the experiment safely, analysing results, and presenting like a real researcher.

WHAT STUDENTS WALK AWAY WITH

Deliverables

01

A research question

A focused, testable hypothesis they can defend.

02

Experiment & data

A clean protocol, real measurements, and graphs.

03

Poster & talk

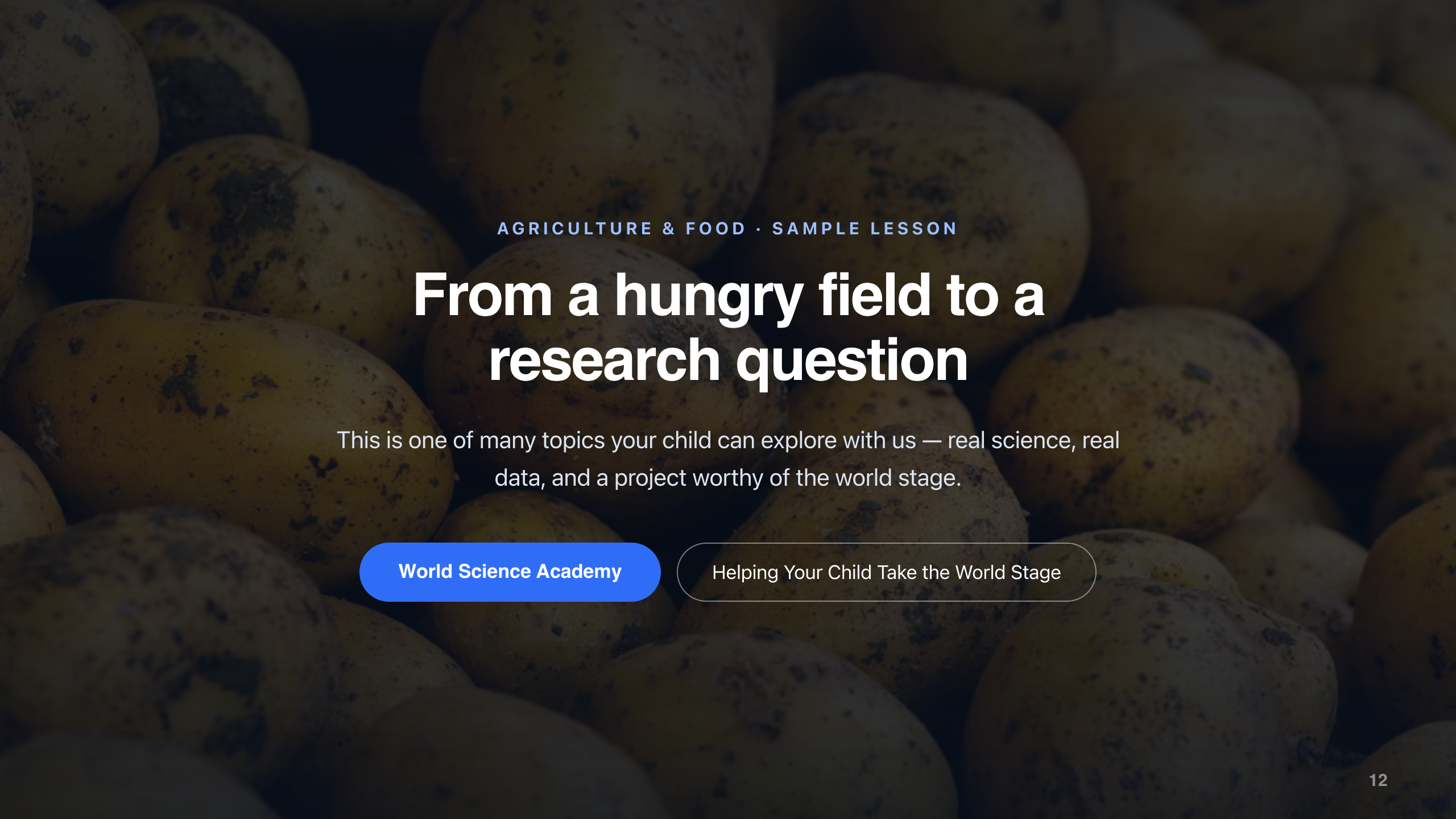
Competition-ready board and a confident presentation.

Scientific writing

An abstract and report in real academic structure.

Judge-ready Q&A

Practice fielding tough questions like a researcher.

The background of the entire slide is a close-up photograph of many potatoes. The potatoes are light brown with dark brown spots (eyes) and are piled together, filling the frame. The lighting is soft, highlighting the texture of the potato skins.

AGRICULTURE & FOOD · SAMPLE LESSON

From a hungry field to a research question

This is one of many topics your child can explore with us — real science, real data, and a project worthy of the world stage.

World Science Academy

Helping Your Child Take the World Stage